Homework - II

Statistics III : Multivariate Data and Regression

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Question 1

Consider a mutiple linear regression model with p predictors and an intercept. Assume that the errors are independent with mean zero and variance σ^2. Let $\hat{y}_i$ be the fitted value of the response Y for the i-th observation with $i = 1, \ldots, n$.

(a) Find the value of $\sum_{i=1}^{n} \text{Var}(\hat{y}_i)$ in terms of n, p and σ^2. Show all your derivation.

(b) State any additional assumptions that you may need.

(c) What does this result say about the impact of including more predictors in the model?

Solution 1

(a) Derivation of $\sum_{i=1}^{n} \text{Var}(\hat{y}_i)$

In matrix notation, the multiple linear regression model is written as

$$Y = X\beta + \epsilon, \qquad E(\epsilon) = 0, \quad \text{Var}(\epsilon) = \sigma^2 I_n,$$

where X is an $n \times (p+1)$ design matrix (including the intercept column) and β is a $(p+1) \times 1$ vector of parameters. The fitted values are given by $\hat{Y} = X\hat{\beta}$. Since $\hat{\beta} = (X^T X)^{-1} X^T Y$, we can write $\hat{Y}$ as:

$$\hat{Y} = X(X^T X)^{-1} X^T Y = HY,$$

where $H = X(X^T X)^{-1} X^T$ is the $n \times n$ hat matrix. Now, we'll calculate the variance-covariance matrix of the fitted values:

$$\begin{aligned}
\text{Var}(\hat{Y}) &= \text{Var}(HY) \\
&= H\text{Var}(Y)H^T \\
&= H(\sigma^2 I_n)H^T \\
&= \sigma^2 HH^T.
\end{aligned}$$

Since H is a symmetric and idempotent matrix ($H = H^T$ and $H^2 = H$), we have $HH^T = H^2 = H$. Thus,

$$\text{Var}(\hat{Y}) = \sigma^2 H.$$

The variance of the i-th fitted value, $\text{Var}(\hat{y}_i)$, is simply the i-th diagonal element of $\sigma^2 H$, which is $\sigma^2 h_{ii}$. The sum of these variances is the trace of the matrix $\sigma^2 H$:

$$\begin{aligned}
\sum_{i=1}^{n} \text{Var}(\hat{y}_i) &= \text{tr}(\text{Var}(\hat{Y})) \\
&= \sigma^2 \text{tr}(H) \\
&= \sigma^2 \text{tr}\big(X(X^T X)^{-1} X^T\big).
\end{aligned}$$

Using the cyclic property of the trace ($\text{tr}(ABC) = \text{tr}(BCA)$), we get

$$\begin{aligned}
\sum_{i=1}^{n} \text{Var}(\hat{y}_i) &= \sigma^2 \text{tr}\big((X^T X)^{-1} X^T X\big) \\
&= \sigma^2 \text{tr}(I_{p+1}),
\end{aligned}$$

where I_{p+1} is the $(p+1) \times (p+1)$ identity matrix. Since the trace of an identity matrix is its dimension,

$$\boxed{\sum_{i=1}^{n} \text{Var}(\hat{y}_i) = \sigma^2(p+1).}$$

(b) Additional assumptions For this derivation to hold exactly in terms of the specific observed X, we assume that the design matrix X is fixed (non-random) and has full column rank, meaning $\text{rank}(X) = p+1$. This ensures that the inverse $(X^T X)^{-1}$ exists. Additionally, we are assuming homoscedasticity (constant variance σ^2) and uncorrelated errors, which was partially given but is formalized as $\text{Var}(\epsilon) = \sigma^2 I_n$.
(c) Impact of including more predictors The sum of the variances of the fitted values is proportional to $(p+1)$. Consequently, the average prediction variance is $\sigma^2(p+1)/n$. This tells us that as we include more predictors (increasing p), the variance of our fitted values linearly increases. Hence, we can conclude that while adding predictors might reduce bias on the training set, it simultaneously inflates the variance of our predictions, increasing the risk of overfitting.

Question 2

Consider simple linear regression and the problem of testing $\beta_1 = 0$. We can do this in two ways:

A. Use a t-test based on $\hat{\beta}_1 \sim \mathcal{N}(\beta_1, \sigma^2/S_{XX})$.

B. Use the F test in the ANOVA.

Answer the following questions in this context.

(a) Write down an estimator for σ^2 that is independent of $\hat{\beta}_1$.

(b) What is the distribution of the estimator in part (a)?

(c) Write down the t-statistic mentioned in (A) and its distribution.

(d) Show that the F-statistic in (B) is $\frac{S_{YY} - \sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2/(n-2)}$.

(e) Show that the numerator in the F-statistic equals S_{XY}^2/S_{XX}.

(f) Hence show that the F-statistic in (B) is the square of the t-statistic in (A).

(g) Conclude that the two tests (A) and (B) are equivalent by expressing the t and F distributions in terms of independent normals.

Solution 2

(a) An unbiased estimator for the error variance σ^2, which is independent of the slope estimator $\hat{\beta}_1$, is the Mean Square Residuals (MSRes). We denote it as s^2:

$$\boxed{s^2 = \frac{1}{n-2} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 = \frac{SSReges}{n-2}}$$

where $SSRes$ is the Sum of Squares due to Residuals.

(b) Assuming the errors are normally distributed, the normalized version of this estimator follows a Chi-square distribution with $n-2$ degrees of freedom:

$$\boxed{\frac{(n-2)s^2}{\sigma^2} \sim \chi^2_{n-2}}$$

(c) The t-statistic for testing the null hypothesis $H_0 : \beta_1 = 0$ is constructed by standardizing $\hat{\beta}_1$ and replacing the unknown σ with its estimator s. Its distribution under H_0 is a Student's t-distribution with $n-2$ degrees of freedom:

$$\boxed{t = \frac{\hat{\beta}_1}{s/\sqrt{S_{XX}}} \sim t_{n-2}}$$

(d) The F-statistic in ANOVA is the ratio of the Mean Square for Regression (MSReg) to the Mean Square for Residual (MSRes).

$$F = \frac{MSReg}{MSRes} = \frac{SSReg/1}{SSRes/(n-2)}$$

We can write $SST = SSReg + SSRes$. Here, $SST = S_{YY} = \sum_{i=1}^{n}(y_i - \bar{y})^2$ and $SSRes = \sum_{i=1}^{n}(y_i - \hat{y}_i)^2$. Rearranging gives $SSReg = S_{YY} - \sum_{i=1}^{n}(y_i - \hat{y}_i)^2$. Substituting this into the F-statistic formula yields:

$$\boxed{F = \frac{S_{YY} - \sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2/(n-2)}}$$

(e) The numerator of the F-statistic is the Regression Sum of Squares, $SSReg = \sum_{i=1}^{n}(\hat{y}_i - \bar{y})^2$. In simple linear regression, the fitted value is $\hat{y}_i = \bar{y} + \hat{\beta}_1(x_i - \bar{x})$. Substituting this, we get:

$$\begin{aligned} SSReg &= \sum_{i=1}^{n} \left(\bar{y} + \hat{\beta}_1(x_i - \bar{x}) - \bar{y}\right)^2 \\ &= \sum_{i=1}^{n} \left(\hat{\beta}_1(x_i - \bar{x})\right)^2 \\ &= \hat{\beta}_1^2 \sum_{i=1}^{n}(x_i - \bar{x})^2 = \hat{\beta}_1^2 S_{XX}. \end{aligned}$$

Since the least squares estimator is $\hat{\beta}_1 = \frac{S_{XY}}{S_{XX}}$, we substitute this back in:

$$SSReg = \left(\frac{S_{XY}}{S_{XX}}\right)^2 S_{XX} = \frac{S_{XY}^2}{S_{XX}^2} S_{XX} = \frac{S_{XY}^2}{S_{XX}}.$$

Thus, the numerator equals S_{XY}^2/S_{XX}.

(f) Taking the square of the t-statistic obtained in part (c):

$$\begin{aligned} t^2 &= \left(\frac{\hat{\beta}_1}{s/\sqrt{S_{XX}}}\right)^2 \\ &= \frac{\hat{\beta}_1^2 S_{XX}}{s^2} \\ &= \frac{SSReg}{MSRes} \end{aligned}$$

As shown in (d) and (e), $\frac{SSReg}{MSRes}$ is exactly the formula for the F-statistic. Therefore,

$$\boxed{t^2 = F}$$

(g) A t-distribution with k degrees of freedom is defined as the ratio of a standard normal random variable Z to the square root of an independent Chi-square random variable U divided by its degrees of freedom: $t_k = \frac{Z}{\sqrt{U/k}}$.

Squaring this definition gives $t_k^2 = \frac{Z^2}{U/k}$. Since $Z \sim \mathcal{N}(0,1)$, its square Z^2 follows a Chi-square distribution with 1 degree of freedom (χ_1^2). An F-distribution F_{ν_1,ν_2} is defined as the ratio of two independent Chi-square variables, each divided by its respective degrees of freedom. Thus:

$$t_{n-2}^2 = \frac{\chi_1^2/1}{\chi_{n-2}^2/(n-2)} \sim F_{1,n-2}$$

Since Z (related to $\hat{\beta}_1$) and U (related to s^2) are independent normal-based statistics, the squared test statistic t^2 identically follows the F-distribution. Thus, the two tests are strictly equivalent for testing $H_0 : \beta_1 = 0$.

Question 3

Consider the data in the file `seatpos.xlsx` **Data description:** Car drivers like to adjust the seat position for their own comfort. Car designers would find it helpful to know where different drivers will position the seat depending on their size and age. Researchers at the HuMoSim laboratory at the University of Michigan collected data on 38 drivers.

(a) Fit a multiple linear regression of `hipcenter` on all other variables.

(b) Comment on the significance of the coefficients and the R^2.

(c) What do you suspect about collinearity from your comment? Compute the correlation matrix of the predictors to check if your suspicion is correct. Explain your conclusion.

(d) Fit simple linear regression models of `hipcenter` on each of the predictors separately. Which predictor gives the highest R^2? Is it very different from the R^2 you obtained from part (b)? Is the regression coefficient significant?

(e) Summarize your findings.

Solution 3

Fitting the MLR model and analyzing significance

We begin by loading the dataset and fitting a multiple linear regression (MLR) model predicting `hipcenter` based on all available physiological predictors.

Listing 1: Multiple linear regression of hipcenter on all other variables

```
library(readxl)
```

```
seatpos <- read_excel("seatpos.xlsx")
seatpos <- na.omit(seatpos)

mlr_model <- lm(hipcenter ~ Age + Weight + HtShoes + Ht + Seated
    + Arm + Thigh + Leg, data = seatpos)
summary(mlr_model)
```

(b) Significance of the coefficients and R^2

The model yields an R^2 of approximately 0.6866, indicating that about 68.7% of the variance in the drivers' hip center position is explained by the combination of these predictors. This is a reasonably high R^2, suggesting a good overall model fit. However, inspecting the individual p-values of the coefficients reveals a stark contrast:

- `Age`: $p \approx 0.184$
- `Weight`: $p \approx 0.937$
- `HtShoes`: $p \approx 0.784$
- `Ht`: $p \approx 0.953$
- `Seated`: $p \approx 0.888$
- `Arm`: $p \approx 0.736$
- `Thigh`: $p \approx 0.671$
- `Leg`: $p \approx 0.182$

At the standard $\alpha = 0.05$ significance level, **none of the individual predictors are statistically significant**.

(c) Suspicions of collinearity and correlation matrix

The combination of a high overall R^2 alongside highly non-significant individual coefficients is a classic symptom of severe **multicollinearity**. When predictors are highly correlated, it becomes difficult for the model to isolate the independent contribution of each variable, resulting in inflated standard errors and small t-statistics.

To verify this suspicion, we compute the correlation matrix.

Listing 2: Correlation matrix of predictors

```
predictors <- subset(seatpos, select = -hipcenter)
cor_matrix <- cor(predictors)
print(round(cor_matrix, 3))
```

	Age	**Weight**	**HtShoes**	**Ht**	**Seated**	**Arm**	**Thigh**	**Leg**
Age	1.000	0.081	-0.079	-0.090	-0.170	0.360	0.091	-0.042
Weight	0.081	1.000	0.828	0.829	0.776	0.698	0.573	0.784
HtShoes	-0.079	0.828	1.000	0.998	0.930	0.752	0.725	0.908
Ht	-0.090	0.829	0.998	1.000	0.928	0.752	0.735	0.910
Seated	-0.170	0.776	0.930	0.928	1.000	0.625	0.607	0.812
Arm	0.360	0.698	0.752	0.752	0.625	1.000	0.671	0.754
Thigh	0.091	0.573	0.725	0.735	0.607	0.671	1.000	0.650
Leg	-0.042	0.784	0.908	0.910	0.812	0.754	0.650	1.000

Table 1: Pearson correlation matrix of the predictor variables.

The correlation matrix overwhelmingly confirms our suspicion. Many physiological variables are strongly positively correlated. For instance, `Ht` and `HtShoes` have a near-perfect correlation of 0.998. Height (`Ht`) is also heavily correlated with `Seated` height (0.928) and `Leg` length (0.910). This massive overlap in information destabilizes the least squares estimation when all variables are included simultaneously.

Fitting separate SLR models

Listing 3: Simple linear regression for each predictor

```r
predictors_names <- names(predictors)
results <- data.frame(Predictor = character(), R_Squared =
    numeric(), P_Value = numeric())

for (var in predictors_names) {
  formula <- as.formula(paste("hipcenter ~", var))
  model <- lm(formula, data = seatpos)
  summ <- summary(model)
  results <- rbind(results, data.frame(
    Predictor = var,
    R_Squared = summ$r.squared,
    P_Value = coef(summ)[2, 4]
  ))
}
print(results)
```

Predictor	R^2	p**-value**
Age	0.0421	2.166×10^{-1}
Weight	0.4100	1.493×10^{-5}
HtShoes	0.6346	2.207×10^{-9}
Ht	**0.6383**	$\mathbf{1.831 \times 10^{-9}}$
Seated	0.5347	1.845×10^{-7}
Arm	0.3423	1.142×10^{-4}
Thigh	0.3495	9.290×10^{-5}
Leg	0.6196	4.587×10^{-9}

Table 2: Results of SLR models for each predictor.

(d) The predictor that gives the highest R^2 is `Ht` (height without shoes), producing an R^2 of 0.6383. This R^2 is remarkably close to the MLR R^2 of 0.6866. Unlike in the MLR, the regression coefficient for `Ht` in this simple model is highly significant ($p = 1.831 \times 10^{-9}$).

(e) Summary of findings

The physiological measurements in this dataset are tightly intertwined. Taller drivers naturally tend to weigh more, have longer legs, arms, and seated heights and hence adding all these features into a single model creates severe multicollinearity. This hides the statistical significance of individual predictors despite the model predicting the `hipcenter` reasonably well as a whole. A simpler model relying on a single robust proxy for overall size, such as bare height (`Ht`), performs almost as well as the complex model while maintaining high statistical significance and offering far better interpretability.

Question 4

Consider the data in the file `earthquake.xlsx`
Data description: The `earthquake` data frame contains measurements of latitude, longitude, focal depth and magnitude for all earthquakes having magnitude greater than 5.8 between 1964 and 1985.
Produce the diagnostic plots of residuals for the multiple linear regression of magnitude using all other variables as predictors. Comment on the plots.

Solution 4

Consider the following

Listing 4: Diagnostic plots for earthquake regression

```r
library(readxl)
earthquake <- read_excel("earthquake.xlsx")
earthquake <- na.omit(earthquake)

eq_model <- lm(magnitude ~ depth + latitude + longitude, data =
    earthquake)

par(mfrow = c(2, 2))
plot(eq_model)
```

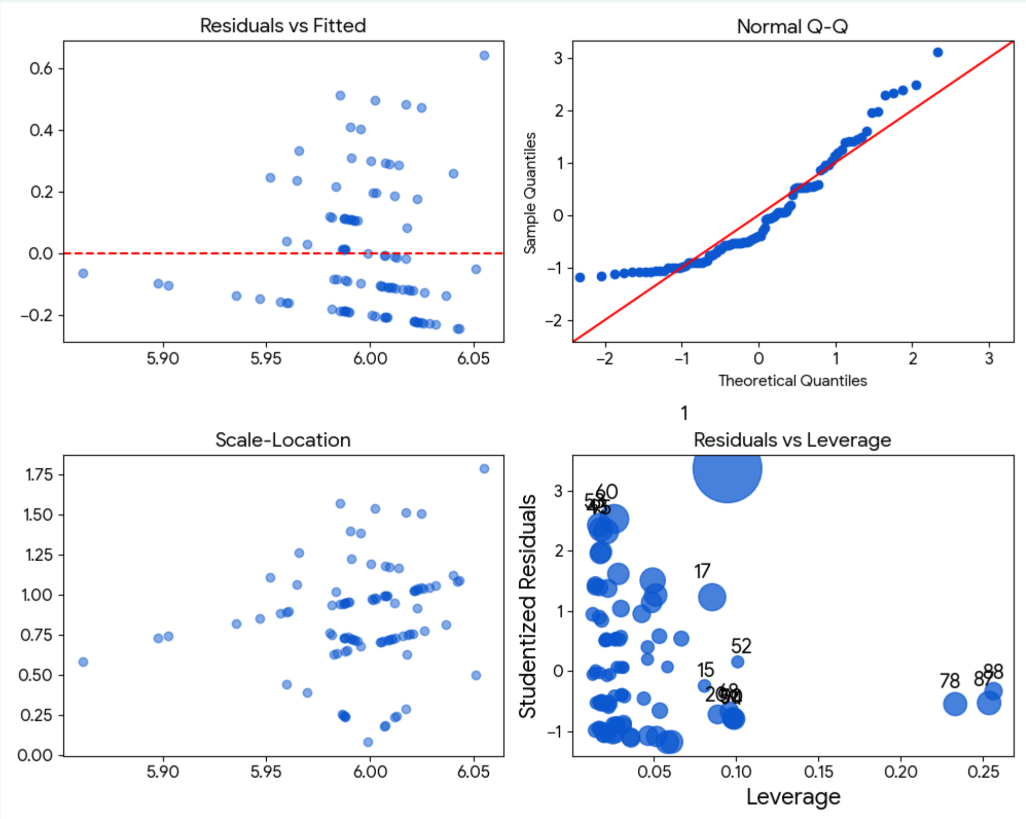

Figure 1: Standard diagnostic plots of residuals for the earthquake MLR model.

Comment on the plots

- **Residuals vs Fitted:** Horizontal banding shows the dependent variable (magnitude) is discretely measured, not continuous.
- **Normal Q-Q:** A heavy right tail indicates non-normal residuals; the model consistently under-predicts high-magnitude earthquakes.
- **Scale-Location:** Confirms the discrete banding, but shows variance remains relatively constant (no severe homoscedasticity violation).
- **Residuals vs Leverage:** No single point exerts undue influence on the model; the sole outlier has low leverage.

Overall: Multiple linear regression is likely unsuitable for this dataset due to the discrete measurements and non-normal residuals.